Abraham-Silberschatz-Operating-System-Concepts-10th-2018

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Ø=ÜË Topics in This Document

- OPERATING
- SYSTEM
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- CONCEPTS
- ABRAHAM SILBERSCHATZ
- PETER BAER GALVIN
- GREG GAGNE
- Chapter 1: Introductionincludes updated coverage of multicore systems,
- Chapter 2: Operating-System Structuresprovides a significantly revised
- Chapter 3: Processessimplifies the discussion of scheduling so that it

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Ø=ÜÝ Section-by-Section Notes

1. OPERATING

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- $BNCSJEHF $PNQVUFS BOE 4UBSGJTI 4UPSBHF
- FTUNJOTUFS $PMMFHF
- (17+ (',7,21
- Publisher Laurie Rosatone

2. 4.2Multicore Programming163

- single core

- Figure 4.3Concurrent execution on a single-core system.
- means that some threads can run in parallel, because the system can assign a
- separate thread to each core (Figure 4.4).
- Notice the distinction betweenconcurrencyandparallelismin this discus-
- sion. Aconcurrent system supports more than one task by allowing all the tasks
- to make progress. In contrast, a parallel system can perform more than one task
- simultaneously. Thus, it is possible to have concurrency without parallelism.

3. 8.6Deadlock Avoidance333

- threads had finished and released its resources, then we could have avoided
- the deadlock.
- Given the concept of a safe state, we can define avoidance algorithms that
- ensure that the system will never deadlock. The idea is simply to ensure that the
- system will always remain in a safe state. Initially, the system is in a safe state.
- Whenever a thread requests a resource that is currently available, the system
- must decide whether the resource can be allocated immediately or the thread
- must wait. The request is granted only if the allocation leaves the system in a

4. 12.5Transforming I/O Requests to Hardware Operations517

- for example, reading a file from disk. The application refers to the data by a
- file name. Within a disk, the file system maps from the file name through the
- file-system directories to obtain the space allocation of the file. For instance, in
- MS-DOSforFAT(a relatively simple operating and file system still used today

- as a common interchange format), the name maps to a number that indicates
- an entry in the file-access table, and that table entry tells which disk blocks
- are allocated to the file. In
- UNIX, the name maps to an inode number, and the

5. 722Chapter 18Virtual Machines

- efficient option. If memory pressure within the entire system decreases,
- VMMwill tell the balloon process within the guest to unpin and free
- some or all of the memory, allowing the guest more pages for its use.
- Another common method for reducing memory pressure is for the
- to determine if the same page has been loaded more than once. If this
- is the case, the
- VMMreduces the number of copies of the page to one
- and maps the other users of the page to that one copy.

6. 6Appendix BWindows 7

- also have been compiled for a different instruction set. Windows 7 implements
- several strategies to run applications despite incompatibilities.
- Like Windows
- XP, Windows 7 has a compatibility layer that sits between
- applications and the Win
- APIs. This layer makes Windows 7 look (almost)
- bug-for-bug compatible with previous versions of Windows. Windows 7, like
- NTreleases, maintains support for running many 16-bit applications

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Ø=Ý Key Definitions

The interrupt routine: indirectly through the table, with no intermediate routine needed

**e handlers on the

corresponding list:** one by one, until one is found that can service the request

A queue in Linux: a
kfifo, and its implementation can be found in thekfifo

This function: when the module is loaded

function: repeatedly until it returns 0, so there must be logic to ensure that this function returns 0 once it has collected the data (in this case, the string "Hello
World n") that is to go into the correspo

**ber of processes
currently in memory:** thedegree of multiprogramming

**This
task:** acontext switchand is illustrated in Figure 3

**and the new
processes:** the children of that process

primitives: follows:

send(P, message) Send amessageto processP

passing facility in Windows: theadvanced local procedure call(
ALPC) facility

+P Important Facts & Formulas

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- 6008, E-Mail: PERMREQ@WILEY.
- Description: 10th edition.
- Classification: LCC QA76.
- O63 (ebook) | LCC QA76.

- required for an understanding of operating systems are covered in Chapter 1.
- results are covered, but formal proofs are largely omitted.
- why we should expect the result in question to be true.
- The book consists of 21 chapters and 4 appendices.
- chronization Tools, and Chapter 7, Synchronization Examples.
- operating systems Linux and Windows 10.
- 1b) and an extension for a threads library (POSIX.

Ø<ß¯ Key Takeaways

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Ø=ÜÌ Exam Tips

- Review each definition and be able to explain it in your own words
- Focus on the formulas and relationships between concepts
- Practice writing short answers for each major topic
- Create your own examples for abstract concepts